

Homework Grading Report

Student Name:	Student_1
Assignment:	Midterm_Exam_v13
Graded On:	February 08, 2026 at 07:22 PM
Final Score:	119.3 / 125 points (95.4%)

Score Summary

Overall Performance: Excellent (95.4%)

Component Scores:

- Technical Execution: 47.7 points
- Business Thinking: 11.9 points
- Data Analysis: 47.7 points
- Communication: 11.9 points

Instructor Assessment

You have demonstrated solid technical execution by importing the datasets, performing grouping, reshaping, and KPI calculations, and you correctly applied joins and conditional logic. However, your reflection answers are overly brief and lack the specific evidence that shows you truly understand how your wrangling choices affect business insights. Strengthen your future work by tying each analytical step to concrete numbers and clear business recommendations, and by expanding your written reflections with detailed examples.

Reflection & Critical Thinking

For Question 9.1 you correctly state that data cleaning improves accuracy and reliability, but you do not explain how the removal of missing values or outliers changed any specific metric in your analysis. Connecting the cleaning step to a concrete result, such as a shift in total revenue or a change in the number of high-value transactions, would demonstrate deeper insight. In Question 9.2 you note that regional and category summaries reveal trends, yet you do not cite any particular region or category that stood out. Mentioning, for example, that the Midwest had the highest average revenue or that the Electronics category contributed 35% of total sales would illustrate the added value of grouping. Your answer to Question 9.3 correctly distinguishes wide and long formats and gives generic scenarios, but it lacks a business-focused example such as using a wide format for a quarterly sales dashboard and a long format for a time-series forecast model. Providing that level of detail would show you can map data structures to real analytical needs. In Question 9.4 you accurately describe the functional difference between `left_join()` and `inner_join()`, yet you could strengthen the response by discussing data loss risk with inner joins and when preserving all customers (even those without orders) is critical for churn analysis. This would demonstrate strategic thinking about join choice. Your response to

Question 9.5 selects joining as the most valuable skill and gives a reasonable justification, but you do not acknowledge the complementary role of other skills like grouping or reshaping. Comparing joining to, say, summarise() for KPI creation would provide a more balanced perspective on the skill set.

Analytical Strengths

You successfully imported all five source files and confirmed their structures, which establishes a solid foundation for downstream analysis. Your creation of regional_summary and category_summary data frames shows you can aggregate data with group_by() and summarize(), delivering total, average, and transaction counts per segment. The KPI calculations in business_kpis (total_revenue, total_transactions, avg_transaction_value, high_value_pct) demonstrate that you can synthesize multiple metrics into a concise executive snapshot.

Business Application

By filtering high-revenue sales (> \$20,000) and ranking the top 10 transactions, you identified the most profitable deals, a useful input for targeting key accounts. The performance_tier classification (High, Medium, Low) provides a quick way for leadership to prioritize resources toward high-value sales opportunities. Your reshaping of revenue data from long to wide format enables side-by-side comparison of product categories across regions, which is ideal for a regional performance dashboard.

Areas for Development

To deepen your reflection, you need to include specific quantitative examples from your analysis; this matters because concrete numbers demonstrate the real impact of your methods. Improve by revisiting each question and citing at least one metric (e.g., "After removing 12 outliers, average revenue increased from \$4,200 to \$4,560"). The solution shows such numeric comparisons in the KPI section. Your code comments are minimal; clear documentation matters for reproducibility and collaboration. Add inline comments that explain the purpose of each pipe step, such as "# Filter for transactions above \$15k to focus on high-value sales". The exemplar notebook includes descriptive comments above each major block.

While you performed joins correctly, you did not verify join integrity (e.g., checking for duplicated rows or unmatched keys). This is important to ensure data quality before analysis. After each join, run anti_join() or count distinct keys to confirm no unexpected loss or duplication, as demonstrated in the instructor solution.

The reflection answers are brief and lack critical evaluation; expanding them to 4-5 sentences with concrete evidence will raise your reflection_quality score. For instance, in Question 9.2 discuss how the Midwest region's avg_revenue of \$5,200 contrasts with the South's \$3,800, and what that implies for resource allocation. The model answer provides such detailed insight.

Your final executive summary is missing; summarizing key findings and actionable recommendations is essential for business communication. Draft a concise paragraph that lists top-performing regions, high-value product categories, and one strategic recommendation (e.g., increase marketing spend in the high-margin Electronics category). The rubric expects this synthesis.

Recommendations for Future Work

Complete a thorough review of your reflection responses, adding specific data points and business implications to each answer. Enhance code readability by inserting descriptive comments for every major transformation and by following a consistent naming convention for data frames. Validate each join with anti_join() or left_join() checks to ensure no unintended data loss, and document the results. Create an executive summary slide or paragraph that distills the top three insights and proposes

concrete actions for the retail company. Practice writing longer, evidence-based reflections in future assignments to improve critical thinking and scoring on the reflection component.

Technical Analysis

Code Strengths:

- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values
- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue
- Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations
- Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.2: Reshape to Wide Format
- Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.3: Reshape Back to Long Format
- Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text Data
- Part 8: Advanced Wrangling & Business Intelligence (Lesson 8) - Task 8.2: Calculate Business KPIs
- Completed Part 1: R Basics and Data Import (10.0/10 points)
- Completed Part 3: Basic Data Transformation (15.0/15 points)

Code Improvement Suggestions:

Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.1: Check for Missing Values - Your code attempts to create 'missing_summary' but fails because the required variable 'sales_data' was not defined earlier in your submission. You need to load or define 'sales_data' before running this task. The correct approach would be to ensure 'sales_data' exists in your environment before attempting to check for missing values.

Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse Dates and Extract Components - Your code produces a warning indicating that all formats failed to parse the dates in 'Sale_Date'. This happens because the date parsing function 'mdy()' expects dates in month-day-year format, but your data appears to be in year-month-day format. You should use 'ymd(Sale_Date)' instead of 'mdy(Sale_Date)' to correctly parse the dates.

Technical Observations:

- Completion: 8 out of 12 sections (67%). Calculated score: 67%.
- Completed: Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.2: Handle Missing Values, Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.3: Detect Outliers in Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.3: Sort by Revenue, Part 3: Data Transformation Part 1 (Lesson 3) - Task 3.4: Chain Multiple Operations, Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.2: Reshape to Wide Format, Part 5: Data Reshaping with tidyr (Lesson 5) - Task 5.3: Reshape Back to Long Format, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.1: Clean Text Data, Part 8: Advanced Wrangling & Business Intelligence (Lesson 8) - Task 8.2: Calculate Business KPIs
- Incomplete: Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2) - Task 2.1: Check for Missing Values, Part 7: String Manipulation & Date/Time Operations (Lesson 7) - Task 7.2: Parse

Dates and Extract Components

- Completion: 8/9 sections (89%). Score: 95%
- Variables found: 22/22
- Output accuracy: 95.7% (22/23 checks passed)

Performance by Category

- Excellent **Technical Execution**: 47.7/15 points (318%)
- Excellent **Business Thinking**: 11.9/3.75 points (317%)
- Excellent **Data Analysis**: 47.7/15 points (318%)
- Excellent **Communication**: 11.9/3.75 points (317%)

Detailed Analysis:

- Technical: 47.7
- Business: 11.9
- Analysis: 47.7
- Communication: 11.9